

## Climate and Precipitation

Canada is, probably more than most countries, affected by its latitude. Canada is a land of extremes. It extends from the warm southern regions at latitudes of 42 degrees to the frigid northern polar cap. It has both humid ocean coasts and dry prairie. It reaches from sea level to the top of Mt. Logan at almost 6 000 m. It experiences four rather distinct seasons of spring, summer, autumn and winter with accompanying weather patterns.

How often do we complain it is too hot, too cold, too wet? These are all results of weather. *Weather* is defined by day-to-day atmospheric conditions of temperature, cloud cover, precipitation, wind direction and speed, humidity and air pressure. The combination of these conditions in certain ways causes *weather patterns*. For example, we know that when a mass of warm moist air runs into a mass of cold dry air it forms a *front* and we will get precipitation as rain, snow, hail or sleet. The long-term weather patterns over an area are what define its *climate*.

Climate is usually referred to as being either *maritime* or *continental*. Maritime climate is found near large water bodies which moderate or keep the temperature from being extremely cold in winter or hot in summer. To explain *maritime climate* we can look at several laws of physics, which come into play here. First, *bright shiny objects reflect heat while dark objects absorb it*. Since large bodies of water reflect the sun, they absorb heat more slowly than the surrounding land, which absorbs it more quickly. Notice how there is usually a cool breeze near water on a summer day. This is due to a law of physics that says *warm air rises cold air falls*. As the warm air over the land rises, it is replaced by the cooler air from over the water. As the land cools more quickly at night, the process is reversed and the warm breeze blows out toward the water. Another law of physics says that *when two objects of different temperatures are put together, they try and find a common temperature*. In this manner, the warm and cold air mix to keep the temperature from changing drastically from summer to winter or day to night. Because oceans are such large bodies of water, they are slow to change temperature and effectively heat the surrounding area in winter and cool it in summer. As well, the air absorbs a large amount of water making this a humid climate. Therefore, maritime climate is found in areas bordering oceans and, to a lesser extent, around very large lakes.

*Continental climate* refers to inland areas, away from moderating water bodies that are extensively influenced by the large landmass absorbing heat from the sun. The land temperature directly influences the air above these areas. If the landmass is hot, then the air above it will be hot. When the sun goes down, the landmass cools quickly and so does the air temperature. In these areas, not only are the winters colder and the summers warmer, but also there is a significant difference between the daytime and night time temperatures. The continental climate is usually much drier than the maritime climate and precipitation is much less.

By the same process that warm and cold air masses move across our land, warm and cold water masses move across the ocean. Large warm bodies of water that are heated by the sun move northward to mix with the large cold bodies of water from the polar regions rushing south. These moving bodies of water create ocean currents. The air above these currents is either warmed or cooled by the water temperature. Warm water from the Pacific, known as the North Pacific Current, runs up the West Coast of Canada and warms the air above it. This causes the warmer, though wetter Pacific climate. On Canada's East Coast, the Gulf Stream brings warm air from the equatorial Atlantic past Nova Scotia and southern Newfoundland to become the North Atlantic Drift past Greenland. In the opposite direction, cold water of the Labrador Current moves down along the coast of Labrador and northern Newfoundland. Where these two air masses meet off eastern Newfoundland they cause large amounts of precipitation and extensive fog banks.

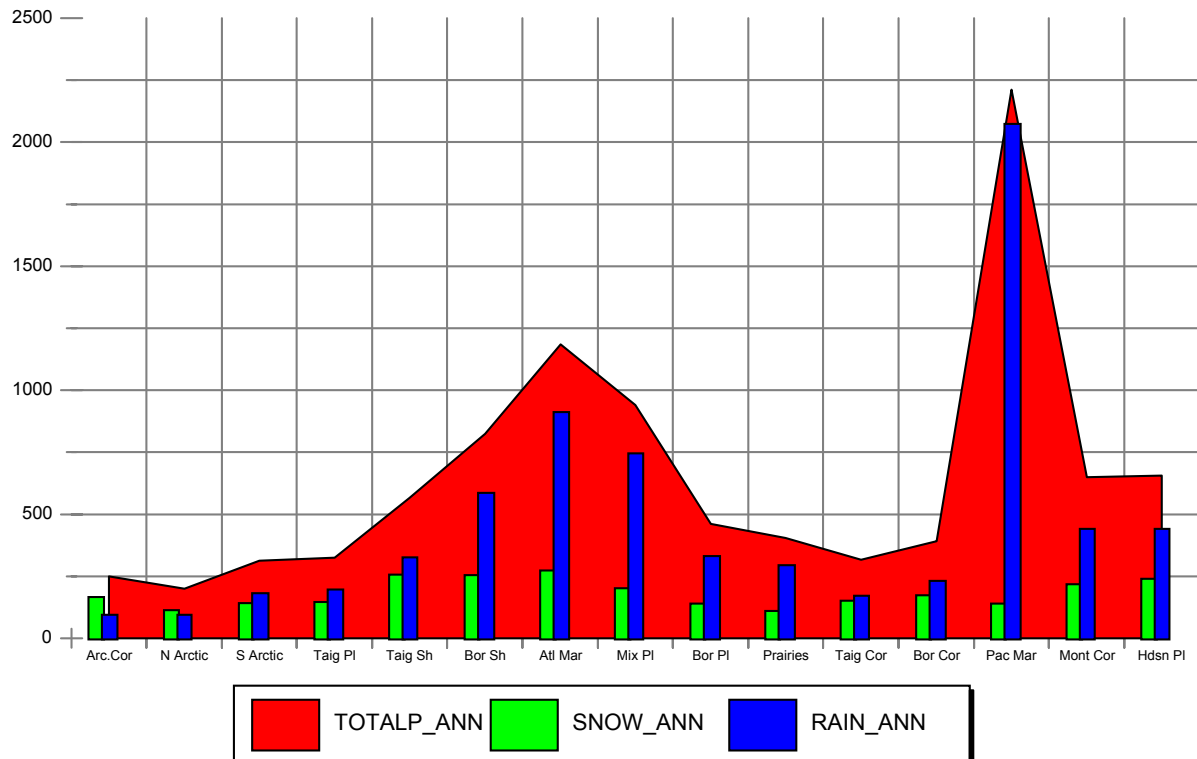
Let's look at some Canadian figures for annual precipitation.

**Table 1: Annual Values for Precipitation (mm) and Temperature (C<sup>0</sup>)**

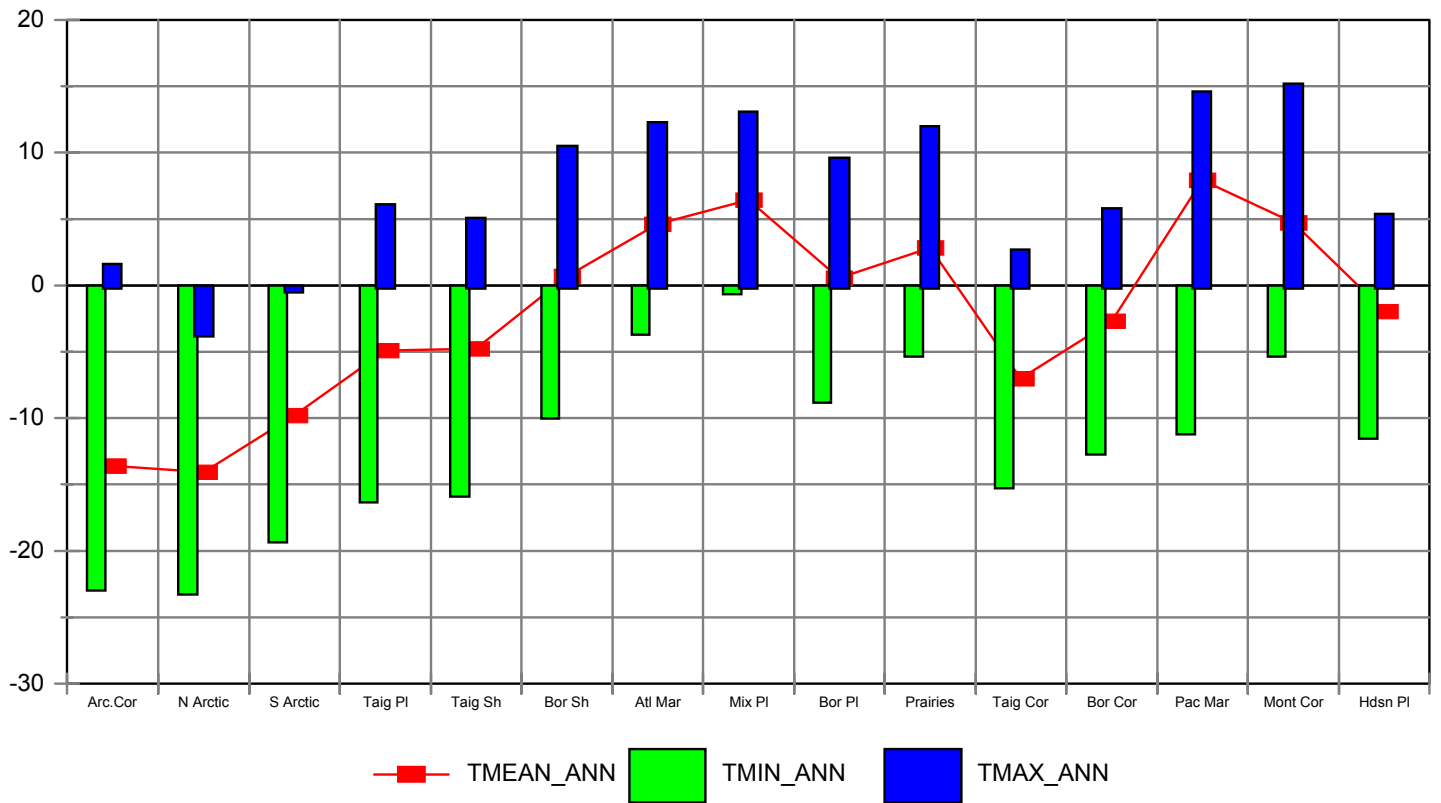
| Ecozone names      | total precip. | snow | rain | min. temp. | max. temp. | mean temp. |
|--------------------|---------------|------|------|------------|------------|------------|
| Arctic Cordillera  | 249           | 167  | 95   | -22.8      | 1.6        | -13.6      |
| Northern Arctic    | 200           | 113  | 94   | -23.2      | -3.7       | -14.0      |
| Southern Arctic    | 313           | 141  | 182  | -19.3      | -0.4       | -9.8       |
| Taiga Plains       | 325           | 147  | 196  | -16.2      | 6.1        | -4.9       |
| Taiga Shield       | 566           | 257  | 327  | -15.7      | 5.1        | -4.8       |
| Boreal Shield      | 825           | 255  | 585  | -9.9       | 10.5       | 0.7        |
| Atlantic Maritime  | 1186          | 272  | 912  | -3.6       | 12.3       | 4.6        |
| Mixedwood Plains   | 941           | 201  | 746  | -0.5       | 13.1       | 6.5        |
| Boreal Plains      | 462           | 141  | 331  | -8.7       | 9.6        | 0.6        |
| Prairies           | 404           | 111  | 294  | -5.2       | 12.0       | 2.9        |
| Taiga Cordillera   | 317           | 152  | 171  | -15.2      | 2.7        | -7.0       |
| Boreal Cordillera  | 392           | 172  | 231  | -12.6      | 5.8        | -2.7       |
| Pacific Maritime   | 2212          | 139  | 2073 | -11.1      | 14.6       | 7.9        |
| Montane Cordillera | 649           | 217  | 441  | -5.2       | 15.2       | 4.7        |
| Hudson Plains      | 656           | 240  | 441  | -11.3      | 5.4        | -1.9       |

Another law of physics is that *warm air absorbs more water than cold air*. As air cools, it releases water in the form of precipitation. We have observed this effect on the maritime coast in the form of rain and fog. However, this effect is also observed over mountainous regions. As warm moist air is pushed up the side of a mountain by prevailing winds, it cools and precipitation falls. This cool air, with its moisture now gone, drops down the leeward or sheltered side of the mountain. With no moisture in the air, this area tends to be extremely dry. This is known as *rainshadow*. If one were to travel across Canada from west to east, this pattern would repeat across each mountain ridge.

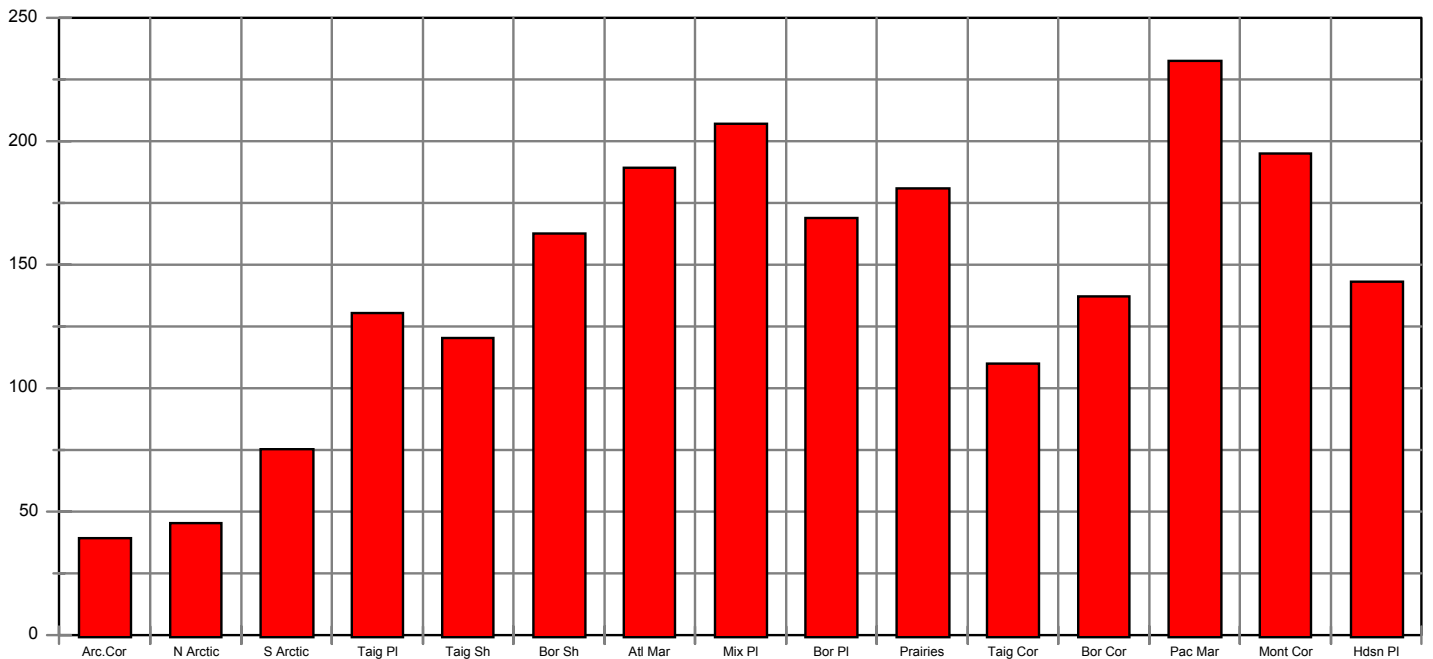
**Chart 1: Using the precipitation, snow and rain values for each ecozone, a graph was drawn below.**



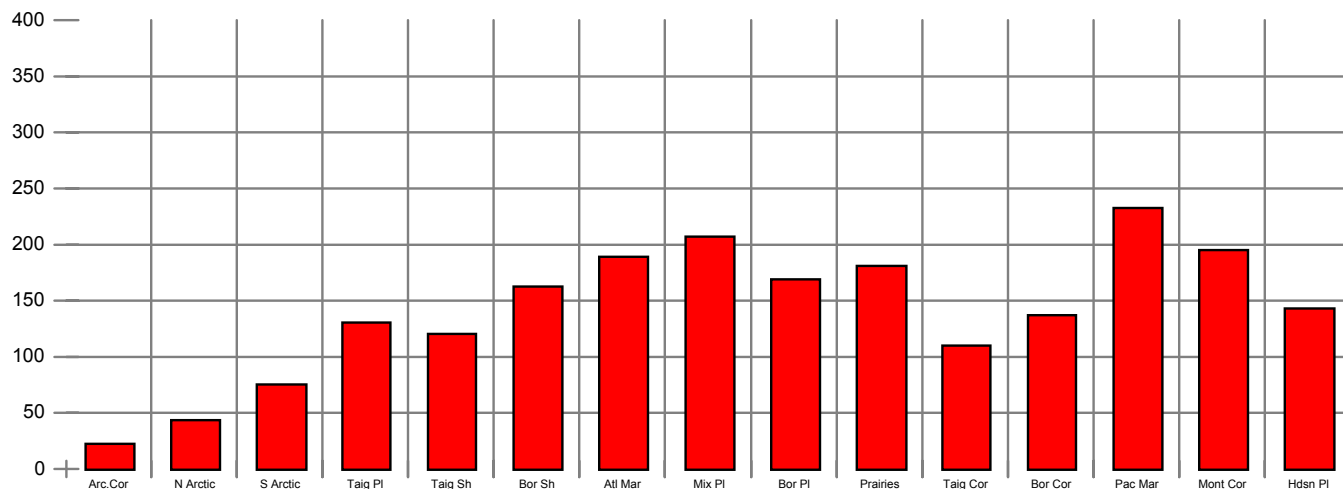
**Chart 2 and 3 (see next page):** Now using the temperature values for the mean, minimum and maximum, let's graph the ecozones. Remember that these temperatures are averaged over the entire ecozone and may not reflect what you might expect knowing the temperature values for a single city or area.



### Number of Growing Days



**Chart 4: Let's look at the number of average growing days in the ecozones (After last frost in spring and before first frost in fall).**



## Climate Change

One of today's major environmental concerns is *climate change*. Change in our climate is occurring for a number of reasons. In the same way that heat is trapped in a greenhouse, the sun's energy is trapped in the earth's atmosphere. The gases, which surround our earth, regulate its temperature by trapping heat from the sun and preventing radiation from escaping our atmosphere. These gases, consisting primarily of water vapour, carbon dioxide, methane and nitrous oxide are known as *greenhouse gases*. These greenhouse gases are important to us, for without them, our average temperature on Earth would drop to well below freezing and we could not exist. However, human activity has produced an excess of greenhouse gases, which is upsetting the natural balance and causing our atmosphere to slowly get warmer.

This "greenhouse effect" is causing concern among scientists. Although the temperature increase over 100 years is just a few degrees, the rate at which greenhouse gases are being produced is rising alarmingly. Greenhouse gases have risen by 30% over the last century, 50% of those increases have occurred in the last 30 years. At this rate, temperatures could increase in some regions by as much as 5 to 10 degrees in the next century. A warmer Canada doesn't sound like such a bad idea, however, there are consequences. Besides changing our annual temperature, climate change affects our weather patterns. It is predicted that we will see more incidents of extreme weather events such as droughts, winter storms, tornadoes and flooding. Consequences of these events would be greater risk to our landscape from erosion; our forests could be damaged by increased pests, diseases and fire; our crops destroyed by drought; our water resources diminished or polluted.